

Project ID	R26-IT-118
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1. Topic (12 words max)

Intelligent Decision Support System for Sri Lankan Tea Estate Management

2. Research group the project belongs to

TIM - Technology Integration and Management
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3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Sri Lanka's tea industry is a major contributor to national export earnings, employment, and rural livelihoods. Despite its global recognition for high-quality tea, operational practices in tea estates and factories remain largely manual, fragmented, and dependent on human judgment. Critical functions such as labor management, climate adaptation, tea leaf disease identification, and tea quality sorting lack integrated, real-time decision support systems. These shortcomings reduce operational efficiency, consistency, and the industry's ability to proactively address environmental and production challenges.

Nature and Gravity of the Problem

The problem affects tea estate workers, supervisors, managers, and factory operators. Managers rely on manual records and subjective supervision, leading to workload imbalance, worker fatigue, and delayed grievance handling. Workers lack structured systems for health and welfare monitoring. Supervisors struggle with timely disease identification due to manual inspection, increasing crop damage risks. Factory operators depend on manual tea sorting, causing inconsistencies, errors, and reduced product quality and efficiency.

Impact of the Problem

Socially, inefficient labor management leads to worker fatigue, reduced job satisfaction, and weakened welfare conditions. Economically, delayed disease detection, climate-related yield losses, inefficient labor use, and inconsistent tea grading reduce profitability and export competitiveness. Technically the absence of integrated intelligent systems limits real-time monitoring and predictive decision making. Environmentally, reactive practices increase chemical use and resource wastage while limited

SDG Alignment

Alignment with Sustainable Development Goals : The research supports SDG 8 through improved labor welfare and productivity, SDG 9 via AI-driven innovation, SDG 12 by ensuring responsible tea production, and SDG 13 through proactive climate risk monitoring and adaptive decision-making.

References

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- [11] J. Blasco, N. Aleixos, and E. Moltó, “Machine vision system for automatic quality grading of fruit,” *Biosyst. Eng.*, vol. 85, no. 4, pp. 415–423, Aug. 2003, doi: 10.1016/S1537-5110(03)00088-6.

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products / Systems

Existing Products and Systems

Existing systems used in tea estates and related agricultural environments provide partial digital support, but their features are limited to specific operational areas.

Labor Management Systems

Sri Lankan tea estates mainly rely on manual attendance registers, spreadsheets, or basic HR software for tracking attendance and wages. International platforms such as AgriWebb (<https://www.agriwebb.com>) and Cropin (<https://www.cropin.com>) offer digital workforce records, task logging, and reporting. However, these systems do not include AI-based fatigue prediction, welfare monitoring, grievance prioritization, or tea estate-specific workforce optimization.

Climate and Weather Monitoring Systems

Weather platforms such as OpenWeather (<https://openweathermap.org>) and WeatherAPI (<https://www.weatherapi.com>), along with data from the Department of Meteorology Sri Lanka (<https://www.meteo.gov.lk>), provide real-time forecasts and historical climate data. These tools supply general weather information but do not generate tea-specific insights such as yield impact, disease risk, or optimal harvesting recommendations.

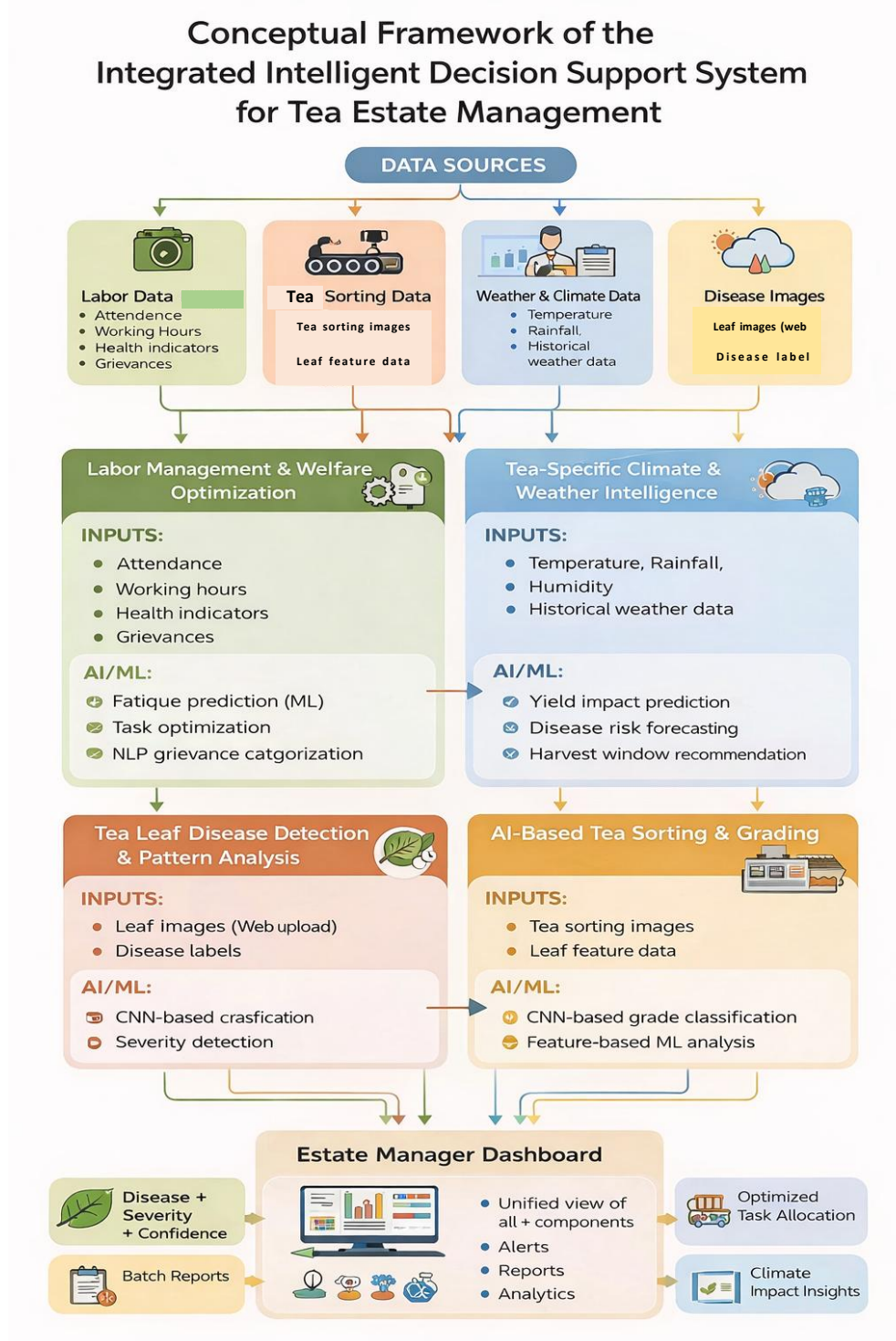
Tea Sorting and Grading Systems

Automated sorting solutions from TOMRA (<https://www.tomra.com>), Bühler (<https://www.buhlergroup.com>), and Satake (<https://www.satake-group.com>) perform high-speed grading based on size, color, and texture. While effective, these systems are expensive, factory-centric, and operate as standalone solutions without estate-level integration.

Existing Research

Academic studies highlight the effectiveness of AI and machine learning in tea estate-related domains. As per Sarker et al. (2021), ML-based workforce management improves productivity through data-driven task allocation, though labor welfare and grievance handling are not fully addressed. Research by Santhiya G. and Radhakrishnan (2025) and Mohanty (2016) demonstrates that CNN-based models achieve high accuracy in tea leaf and plant disease detection; however, these studies focus mainly on classification and lack operational integration. Tea quality assessment research shows that deep learning improves grading accuracy and consistency (Zhang et al., 2023; Blasco et al., 2003), but such systems are typically factory-centric. Climate-based yield prediction studies (Batoool et al., 2022; Mampitiya et al., 2025) further confirm the value of predictive analytics, yet remain isolated from holistic estate management solutions.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



Description of the Main System Components

● Labor Management & Welfare Optimization

This component focuses on improving workforce efficiency and worker welfare within tea estates. It uses labor related inputs such as attendance records, working hours, health indicators, and grievance submissions collected through the web application. Machine learning models analyse these inputs to detect workload imbalances and predict worker fatigue levels. Based on the predictions, the system recommends optimized task allocations to prevent overburdening and ensure fair workload distribution. In addition, an NLP based grievance categorization mechanism automatically classifies and prioritizes worker grievances, enabling faster response and resolution. The outputs of this component include optimized task schedules, welfare alerts, and grievance priority levels, supporting both productivity improvement and ethical labor management.

● Tea-Specific Climate & Weather Intelligence

This component transforms generic weather data into tea-specific climate intelligence. It processes inputs such as temperature, rainfall, humidity, and historical weather patterns relevant to tea-growing regions. Machine learning models analyse climate trends to predict yield impacts, assess disease risk probabilities, and identify optimal harvesting windows. Unlike conventional weather applications, this module provides actionable insights tailored to tea cultivation stages and estate conditions. The outputs include climate alerts, yield forecasts, and harvest recommendations, allowing estate managers to proactively respond to climate variability, minimize losses, and enhance crop quality.

● Tea Leaf Disease Detection & Pattern Analysis

The disease detection component supports early identification and management of tea leaf diseases. It uses leaf images uploaded through the web application along with disease labels verified by supervisors or experts. A CNN based model classifies diseases and determines severity levels, while historical disease records are analysed to identify recurring patterns and seasonal trends. This component not only provides instant disease diagnosis but also supports long-term disease monitoring and prevention strategies. The outputs include detected disease type, severity level, treatment recommendations, and disease trend reports, enabling timely intervention and reduced crop damage.

● AI-Based Tea Sorting & Grading

This component automates tea quality assessment at the factory level. It processes tea sorting images and extracted leaf feature data such as size, shape, colour intensity, texture, and broken versus whole leaf ratio. CNN-based grade classification combined with feature-based machine learning models categorizes tea into Dust, Pekoe, BOP, and FBOP grades. The system generates confidence scores and batch-wise production reports, ensuring consistency and accuracy in grading. By integrating sorting outputs with estate-level intelligence, this component supports quality control, operational transparency, and improved market competitiveness.

Key Benefits of the System

1. Integrated Decision Support

Combines labor management, climate intelligence, disease detection, and tea sorting into a single AI-enabled platform, eliminating fragmented decision-making and improving overall operational coordination.

2. Improved Labor Welfare and Productivity

AI-based fatigue prediction, task optimization, and grievance prioritization enhance worker welfare, reduce overburden, and improve workforce efficiency within tea estates.

3. Proactive Climate and Risk Management

Tea-specific weather intelligence enables early detection of climate-related risks, accurate yield forecasting, and optimized harvesting decisions, reducing climate-induced losses.

4. Early Disease Detection and Prevention

CNN-based disease identification and pattern analysis support timely intervention, reduce crop damage, and minimize excessive chemical usage, promoting sustainable practices.

5. Consistent and Accurate Tea Grading

Automated tea sorting ensures uniform quality, reduces human error, and improves batch consistency, enhancing market value and export competitiveness.

6. Data-Driven and Sustainable Operations

Real-time dashboards, alerts, and analytics support informed decision-making, resource efficiency, and long term sustainability across tea estate and factory operations.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data.
(500words max)

The proposed research focuses on developing an Integrated Intelligent Decision Support System (I-IDSS) for Sri Lankan tea estate management by integrating Artificial Intelligence (AI), Machine Learning, and web technologies. Although the project team does not possess prior professional experience working within the tea industry, the required domain expertise, technical knowledge, and data availability are ensured through structured industry collaboration, guided supervision, and a clearly planned data collection strategy.

Availability of Domain Expertise and Industry Knowledge

Domain knowledge related to tea estate and factory operations is acquired through direct engagement with the tea industry. The research team plans to formally contact and collaborate with selected tea estates and tea factories to understand real-world operational practices. This includes field visits, interviews, and discussions with estate managers, field supervisors, and factory operators who have hands-on experience in labor management, tea leaf disease identification, climate-sensitive decision-making, and tea sorting processes.

These industry practitioners act as domain mentors, assisting the research team in understanding daily workflows, operational constraints, and decision-making criteria used in tea estates. In addition, a domain supervisor or advisory contact from the tea estate side is expected to support the research by validating assumptions, confirming data relevance, and reviewing system outputs from a practical perspective. This ensures that the proposed system reflects real operational needs rather than theoretical assumptions.

To further strengthen domain expertise, the research relies on institutional and authoritative secondary sources. The Tea Research Institute of Sri Lanka (TRI) provides agro-climatic recommendations, disease management guidelines, and tea processing standards relevant to local conditions (<https://www.tri.lk>). Government publications, estate management manuals, and industry reports are also reviewed to ensure regulatory compliance and contextual accuracy.

Technical Knowledge and Research Skills

The technical knowledge required for the project is developed through academic coursework, supervised research activities, and continuous literature review. The research team applies established AI and machine learning techniques such as CNN-based classification, feature-based machine learning models, and predictive analytics using peer-reviewed IEEE and scientific literature. Web-based system development skills, including dashboards and API integration, are acquired using standard frameworks and academic resources.

Data Collection

The research uses a combination of primary and secondary data, clearly planned for each system component.

Primary Data (Planned Collection)

Primary data will be collected during the research phase through collaboration with selected tea estates and factories

Labor Data: Attendance records, health records, working hours, task allocation details, and grievance records obtained from estate supervisors and manual registers.

Tea Leaf Disease Data: Leaf images and disease labels captured and verified by supervisors using the web-based application.

Tea Sorting Data: Feature-based samples (Dust, Pekoe, BOP, FBOP) collected manually at factory level under supervision.

Secondary Data (Publicly Available Sources)

Secondary datasets support model training and validation

Weather & Climate Data:

OpenWeather API – <https://openweathermap.org/api>

Department of Meteorology Sri Lanka – <https://www.meteo.gov.lk>

Agro-climatic & Tea Guidelines:

Tea Research Institute of Sri Lanka – <https://www.tri.lk>

Reference Datasets & Research:

IEEE Xplore – <https://ieeexplore.ieee.org>

Kaggle (plant disease datasets) – <https://www.kaggle.com>

9. Objectives and Novelty

Main Objective To design and evaluate an Integrated AI-Driven Intelligent Decision Support System for Sri Lankan tea estates that improves labor welfare and productivity, enables tea-specific climate and weather intelligence, supports early tea leaf disease detection using CNN-based image analysis, and automates tea sorting and grading, thereby enhancing operational efficiency, sustainability, and decision-making accuracy across tea estate and factory operations			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Gunasekara L.M.N.P IT22267986	Labor Management & Welfare Optimization	• Collect worker attendance, working hours, health indicators, task performance, and grievance data • Analyze historical labor data to identify workload imbalance and fatigue patterns • Design and train ML models for fatigue prediction • Implement AI-based task allocation recommendations • Develop NLP based grievance categorization and prioritization system • Design estate manager dashboard for labor decision support	Introduces an AI-driven labor welfare management approach, not limited to productivity tracking. Unlike existing labor systems that only record attendance and wages, this component predicts worker fatigue, optimizes task allocation, and automates grievance prioritization using NLP. Integrates worker welfare, health awareness, and rights support into a unified decision support system, which is largely absent in existing tea estate labor management solutions.
Gunasekara G.N.D IT22279620	Tea-Specific Weather and Climate Intelligence System	• Collect and integrate multi source historical and real-time weather data • Develop tea-specific climate interpretation layer • Build ML models for weather pattern recognition and anomaly detection • Correlate climate data with yield, quality, and disease risk • Implement automated alerts and intelligent reminder system • Design annual cultivation and harvesting calendar optimizer	Transforms generic weather data into tea-specific cultivation intelligence. Unlike conventional weather apps that provide raw forecasts, this system uses AI to directly relate climate patterns to tea yield, quality, and disease risk. Provides proactive, estate-specific recommendations without requiring IoT infrastructure, making it cost-effective and suitable for Sri Lankan tea estates.
Miuranga W.A.R IT22209320	Tea Leaf Disease Detection and Pattern Analysis	• Collect and label tea leaf disease images • Pre process images and design CNN architecture • Train and evaluate CNN for disease type and severity prediction • Develop treatment recommendation logic • Analyze historical disease data for seasonal and location-based trends	Creates a tea-specific disease image dataset and applies CNNs tailored to Sri Lankan tea leaf characteristics, unlike existing studies focused on generic crops. Combines real-time disease detection with severity estimation and long-term pattern analysis, enabling predictive disease management rather than isolated diagnosis.

Pathirana I.M IT22282668	AI-Based Tea Sorting and Grading Automation	• Collect and label tea grade images (Dust, Pekoe, BOP, FBOP) from factory environment • Preprocess images under controlled conditions • Train CNN models using size, texture, color, and broken-leaf ratio features • Generate batch-wise grade classification and daily production reports • Compare AI-based grading accuracy with manual sorting	Applies CNN-based tea grading to reduce manual inconsistency and subjectivity. Unlike existing factory-centric sorting systems, this research integrates confidence scoring, batch-level analytics, and performance comparison with manual sorting, focusing on research evaluation and decision support rather than automation alone.
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This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes ☒ No ☐

b) The proposed topic exhibits novelty.

Yes ☒ No ☐

c) The student group can successfully execute the proposed project.

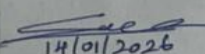
Yes ☒ No ☐

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes ☒ No ☐

e) Any other comments:

Approved.

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Uthpala	Samarakoon	 14/01/2026
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				